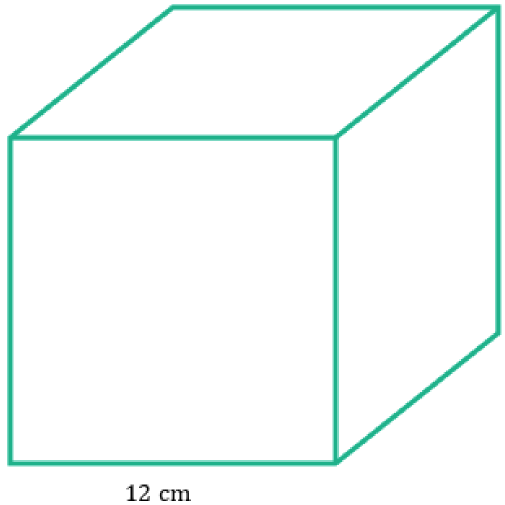
# Volume of prisms and cylinders

1 Find the volume of the cube shown.



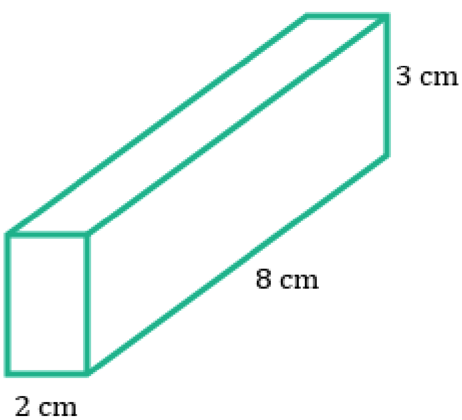

2 Find the side length of a cube with volume $27 \text{ cm}^3$.

3 Find the length of the rectangular prism with volume $162 \text{ cm}^3$, width $6 \text{ cm}$ and height $3 \text{ cm}$.

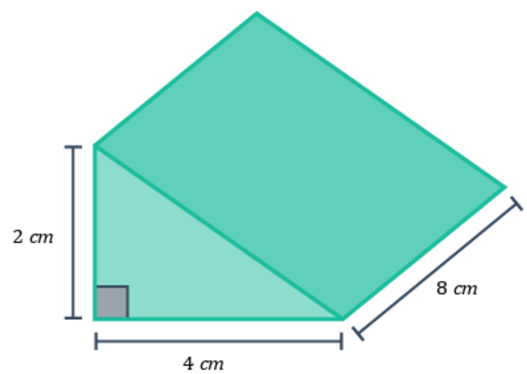
4 A pentagonal prism has a volume of $176.4 \text{ m}^3$ and a cross sectional area of $29.4 \text{ m}^2$.
Find the height of the pentagonal prism.

5 Find the volume of the following prisms:

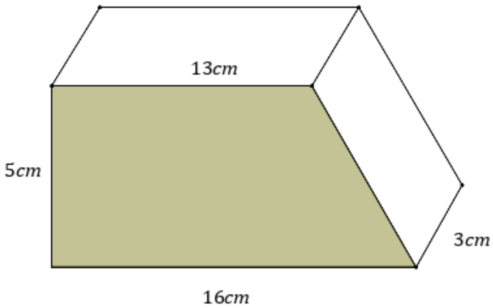
a



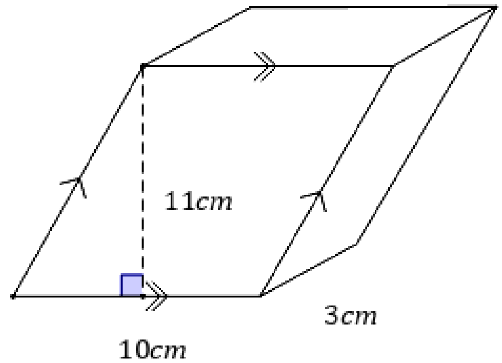

b



c

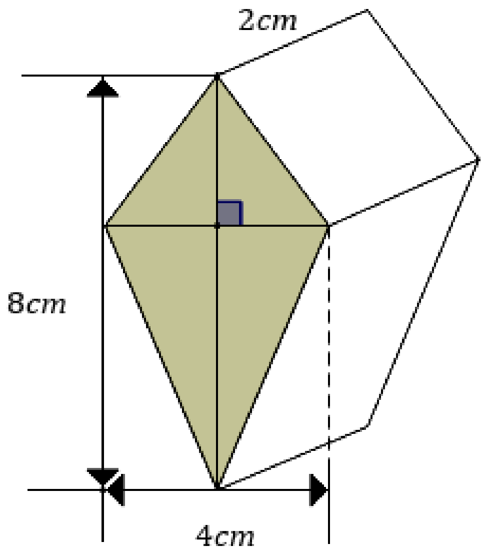


13cm
5cm
3cm
16cm

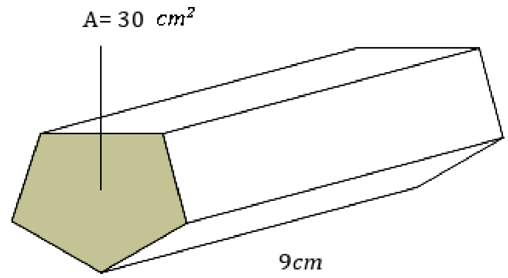
11cm
3cm
10cm


e



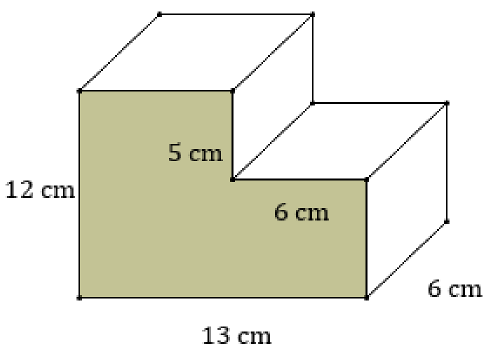
2cm
8cm
4cm


f



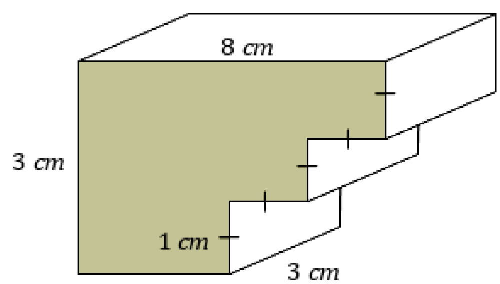
A= 30 cm²
9cm


g



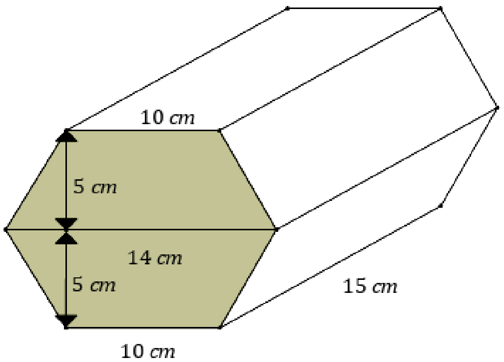
5 cm
12 cm
6 cm
6 cm
13 cm

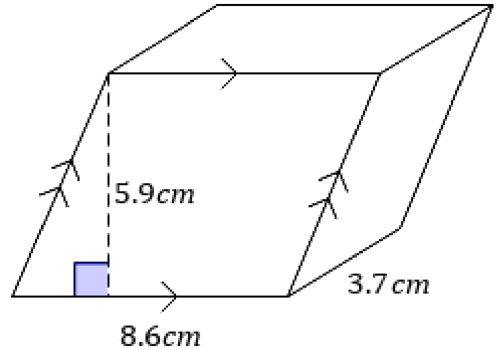

h



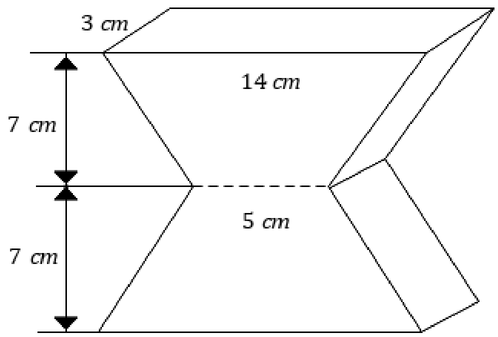
8 cm
3 cm
1 cm
3 cm

i

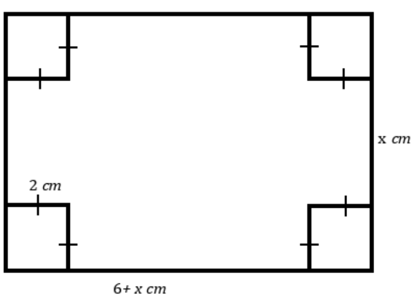



k




**6** The length of a rectangular piece of cardboard is $6 \text{ cm}$ longer than its width. Squares with sides of length $2 \text{ cm}$ are cut from the four corners, and the flaps are folded upward to form an open box with a volume of $32 \text{ cm}^3$.
Determine whether following equations correctly describes the volume of the box.




a $x(x+6) = 32$

b $x(x+6)(2) = 32$

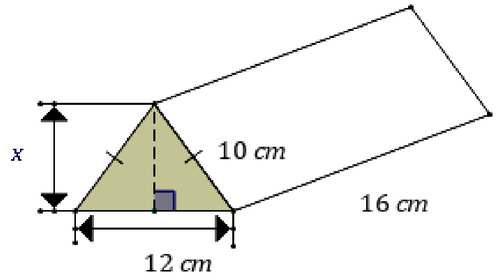
c $(x+2)(x-4) = 32$

d $(x+2)(x-4)(2) = 32$

7 Consider the following diagram.

a Find the value of the unknown height $x$.

b Hence find the volume of the solid.

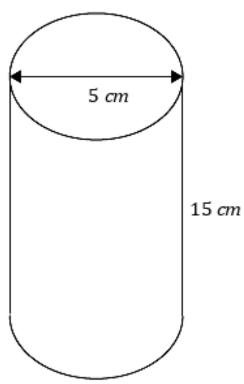



8 Find the volume of a cylinder correct to one decimal place if its diameter is $2$ cm and its height is $19$ cm.

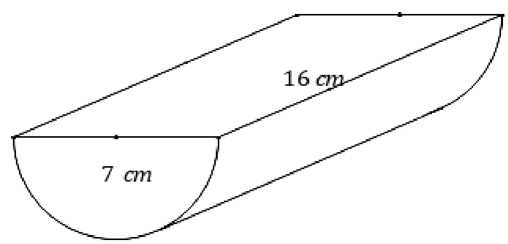
9 Find the volume of a cylinder with radius $7$ cm and height $15$ cm, correct to two decimal places.

10 Find the volume of the following solids, correct to one decimal place. All measurements are in centimeters.

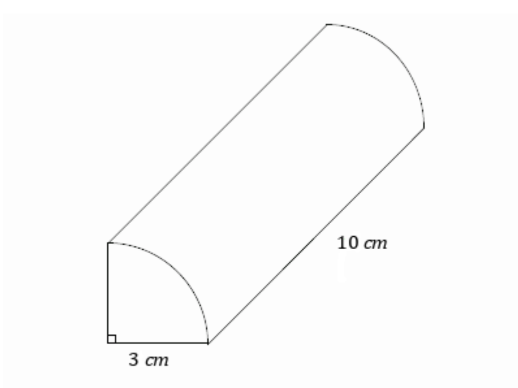
a



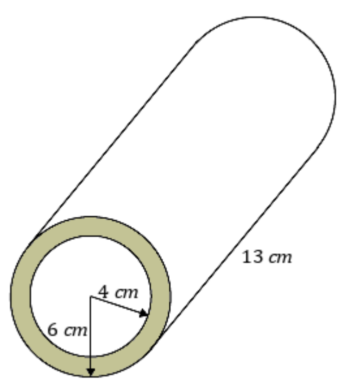

b



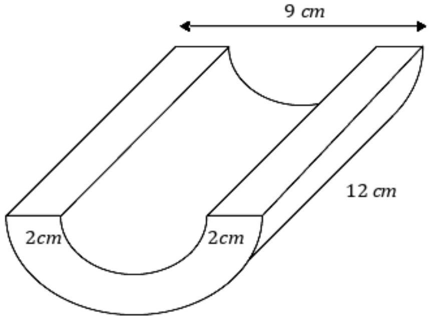

c

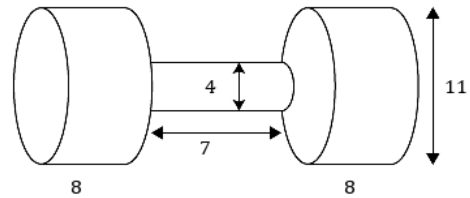



d

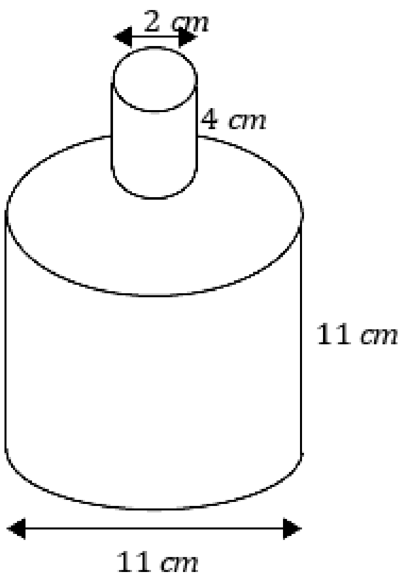


e

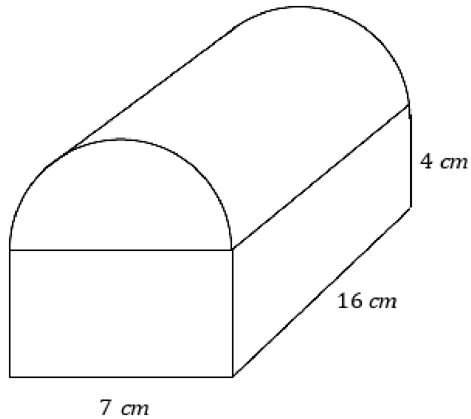

9 cm
12 cm
2cm
2cm


4
11
7
8
8

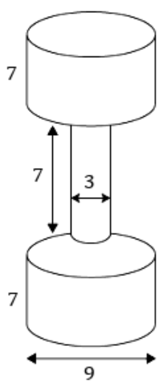
g


2 cm
4 cm
11 cm
11 cm

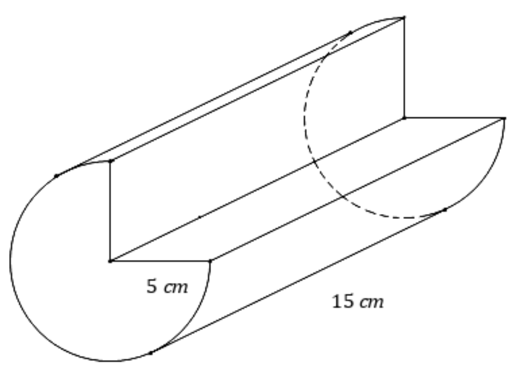
h


4 cm
16 cm
7 cm

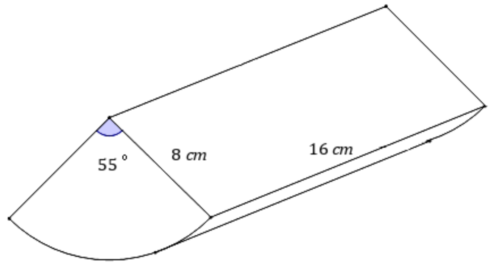
i


7
7
3
7
9

j


5 cm
15 cm

**k**

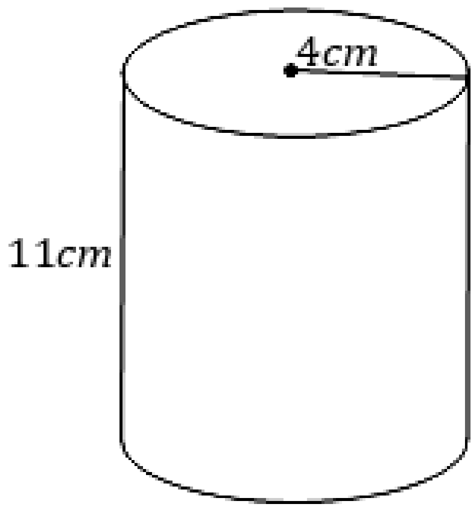



**11** Xavier's mother told him to drink $4$ glasses of water each day. She gave him a cylindrical glass with height $11$ cm and radius $4$ cm.

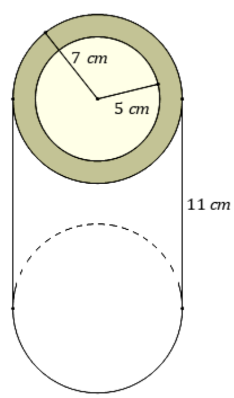
**a** Find the volume of the glass.
Round your answer to two decimal places.

**b** Assuming that he drinks $4$ full glasses as his mother suggested, calculate the volume of water Xavier drinks each day.
Round your answer to two decimal places.

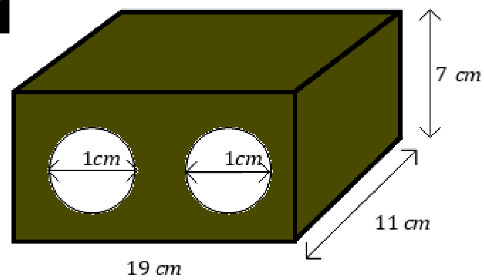
**c** If Xavier follows this drinking routine for a week, how many litres of water would he drink altogether?
Round your answer to one decimal place.




**12** A cylinder is hollowed out by a hole of radius $5$ cm. Find the volume of the remaining figure, correct to two decimal places.



**13** We wish to calculate the volume of the solid shown by considering it as a prism. The cylinders are drilled perpendicular to the surface.



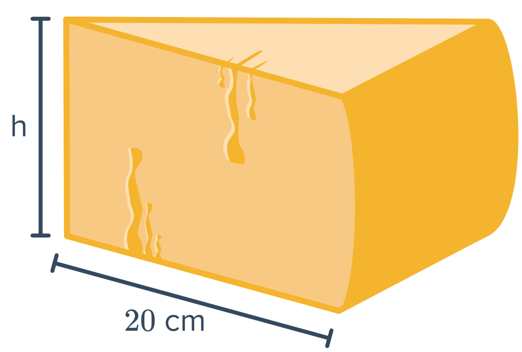

**a** Calculate the area of the front face correct to two decimal places.

**b** Hence find the volume of the prism.

## Additional questions

**14** Queso Fresco is sold in wheels. The volume of a wheel of Queso Fresco is $2002\ \text{cm}^3$: The height of a wheel of Queso Fresco is $6\ \text{cm}$, find the radius.


**15** Parmigiano-Reggiano is formed in wheels, and then cut into twenty equal slices before being packaged and sold. If the volume of a certain wheel of Parmigiano-Reggiano is $8000\pi\ \text{cm}^3$:

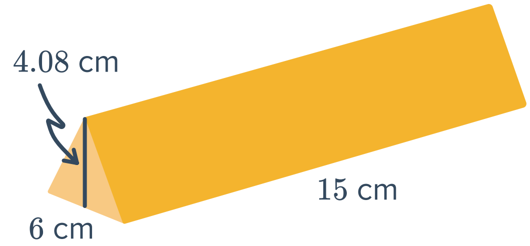



**a** Find the volume of a slice of Parmigiano-Reggiano that has been packaged for sale.

**b** Find the height of the wheel of Parmigiano-Reggiano.

**16** A standard six-sided die has a side length of $5\ \text{mm}$. If Kyle keeps his die in a pouch with a volume of $1000\ \text{mm}^3$, how many die can he fit in the pouch?

17 A chocolate company sells their bars in a special triangular prism packaging:



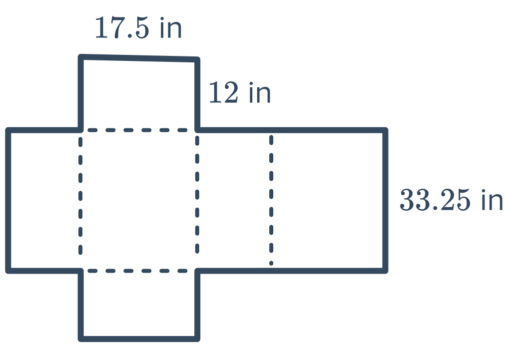

a Find the volume of the chocolate packaging.

b The chocolate only fills $85\%$ of the packaging. Find the volume of one bar of chocolate.

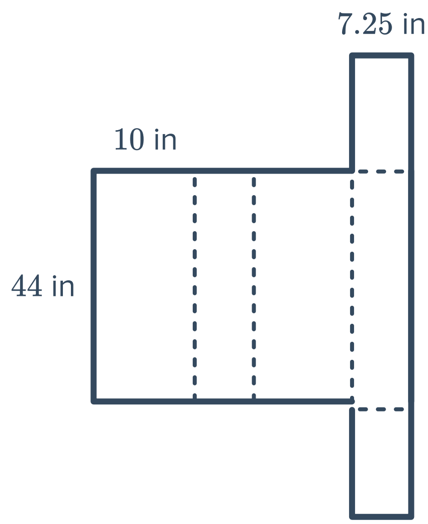
18 Tarpon Springs Aquarium is designing a new Reef Shark exhibit. In order to house the sharks they need a tank with a volume of $128\,000$ gallons.
In order to prevent the salt water from spilling over the edge of the tank, there needs to be a $1.5$ meter clearance from the water level to the top of the shark tank. Using the conversion $1\text{ m}^3 = 244\text{ gal}$, design a rectangular tank that can hold $128\,000$ gallons of salt water.

19 U-Pack Moving Company sells two sizes of moving boxes. U-Pack claims to keep their prices lower than competitors by selling their boxes flat, and having customers assemble the boxes themselves.

Box 1 sells for $\$2.29$:




Box 2 sells for $\$0.99$:




If you have $60\text{ ft}^3$ of stuff to pack into boxes for your move, which of the following combinations of boxes would you buy? Explain.

a $15\times$ Box 1

a $36\times$ Box 2